

Need ID	Description
UN001	must accurately detect cardiac events
UN002	must be able to monitor long term
UN003	must be able to function remotely
UN004	must be comfortable to wear
UN005	must be easy to setup app as well as the device
UN006	must have minimal user interaction after setup
UN007	must be discreet for social setting
UN008	must be compatible with skin (skin irritation, allergic reaction)
UN009	must be wire free for better mobility and comfort
UN010	must be non-intrusive (not interfere with daily lifestyle)
UN011	must be easily replaceable or maintained

Input ID	Description	Linked Need ID
DI001	The device shall use a medical-grade ECG front-end (e.g., ADAS1000) with a sampling rate ≥ 250 Hz and signal-to-noise ratio ≥ 20 dB.	UN001
DI002	The device shall store and/or transmit ECG data continuously for ≥ 7 days with 1-minute resolution.	UN002
DI003	The device shall support Bluetooth Low Energy (BLE 5.0) with automatic data syncing to cloud via mobile app.	UN003
DI004	The patch shall weigh < 25 g and use breathable, hypoallergenic adhesive suitable for continuous wear.	UN004
DI005	The mobile app shall feature an onboarding wizard and video instructions for setup.	UN005
DI006	The device shall automatically start monitoring after setup and require no further input during regular operation.	UN006
DI007	The patch shall have a form factor < 5 mm thick with skin-tone or transparent housing for discreetness.	UN007
DI008	The adhesive and outer casing shall be ISO 10993 biocompatible and dermatologically tested.	UN008
DI009	The patch shall operate wirelessly with internal battery and zero wired components during use.	UN009
DI010	The device shall allow full range of motion (e.g., sitting, walking, sleeping) without mechanical interference.	UN010

DI011

The device battery and adhesive shall be modularly replaceable by user in < 1 minute using a slide-lock mechanism.

UN011

Output ID	Description	Linked Input ID
DO001	ECG signal acquisition validated using ADAS1000 at 250 Hz in test bench and signal-to-noise ratio confirmed ≥ 20 dB.	DI001
DO002	Data logging and transmission verified with 7-day continuous operation in simulated patient environment.	DI002
DO003	BLE 5.0 module integrated with successful data transmission test to Android/iOS app; encryption logs verified.	DI003
DO004	Device weight measured at 22g; adhesive passed 7-day continuous wear comfort test.	DI004
DO005	Mobile app onboarding tested with 10 users; setup completed in < 3 minutes with 95% satisfaction rate.	DI005
DO006	Software confirmed to auto-start monitoring post-setup without manual intervention during 72-hour test cycle.	DI006
DO007	Final patch enclosure is 4.5mm thick with neutral tone; confirmed unnoticeable under clothing in user trial.	DI007
DO008	ISO 10993 biocompatibility reports completed and passed for adhesive and enclosure; no irritation in 10-user patch test.	DI008
DO009	Wireless functionality tested under movement and daily activities; no disconnection over 72-hour dynamic test.	DI009

DO010	Patch tested on users during sitting, walking, and sleeping; motion artifact < 5% in ECG recording.	DI010
DO011	Battery and adhesive module replaced by users in < 1 minute; validated through usability testing.	DI011

Input ID	Verification Method
DI001	Bench test ECG signal using waveform simulator and confirm sampling rate and signal-to-noise ratio meet specs.
DI002	Continuous data logging test in simulated use case for 7 days with battery life monitoring and data integrity check.
DI003	Perform Bluetooth Low Energy (BLE) range test and validate AES encryption via transmission log analysis.
DI004	Measure device weight and conduct human wear test for 7 days with comfort feedback survey.
DI005	Usability test with 10 users setting up device and app; measure completion time and error rate.
DI006	Run 72-hour software test cycle to ensure automatic monitoring starts post-setup with no manual actions.
DI007	Measure patch thickness and conduct visibility test under clothing on a diverse user group.
DI008	Submit adhesive and enclosure for ISO 10993 biocompatibility testing and patch wear irritation test.
DI009	Wireless performance test with daily activity simulation and real-time disconnection logging.
DI010	Motion artifact evaluation during sitting, walking, and sleeping using annotated ECG signal recordings.

DI011

Conduct usability test where users replace battery and adhesive within 1 minute; measure completion time and error rate.

Date	Change Description	Author
2024-06-01	Initial design inputs uploaded	Shivangi
2024-06-10	Updated risk assessment thresholds	QA Engineer
2024-06-15	Added verification test cases	Design Lead

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The device shall support Bluetooth Low Energy (BLE 5.0) with automatic data syncing to cloud via mobile app.	DO003	BLE 5.0 module integrated with successful data transmission test to Android/iOS app; encryption logs verified.
The patch shall weigh < 25 g and use breathable, hypoallergenic adhesive suitable for continuous wear.	DO004	Device weight measured at 22g; adhesive passed 7-day continuous wear comfort test.
The mobile app shall feature an onboarding wizard and video instructions for setup.	DO005	Mobile app onboarding tested with 10 users; setup completed in < 3 minutes with 95% satisfaction rate.
The device shall automatically start monitoring after setup and require no further input during regular operation.	DO006	Software confirmed to auto-start monitoring post-setup without manual intervention during 72-hour test cycle.
The patch shall have a form factor < 5 mm thick with skin-tone or transparent housing for discreetness.	DO007	Final patch enclosure is 4.5mm thick with neutral tone; confirmed unnoticeable under clothing in user trial.
The adhesive and outer casing shall be ISO 10993 biocompatible and dermatologically tested.	DO008	ISO 10993 biocompatibility reports completed and passed for adhesive and enclosure; no irritation in 10-user patch test.
The patch shall operate wirelessly with internal battery and zero wired components during use.	DO009	Wireless functionality tested under movement and daily activities; no disconnection over 72-hour dynamic test.

The device shall allow full range of motion (e.g., sitting, walking, sleeping) without mechanical interference. DO010

Patch tested on users during sitting, walking, and sleeping; motion artifact < 5% in ECG recording.

The device battery and adhesive shall be modularly replaceable by user in < 1 minute using a slide-lock mechanism. DO011

Battery and adhesive module replaced by users in < 1 minute; validated through usability testing.

Verification_Method

Bench test ECG signal using waveform simulator and confirm sampling rate and signal-to-noise ratio meet specs.

Continuous data logging test in simulated use case for 7 days with battery life monitoring and data integrity check.

Perform Bluetooth Low Energy (BLE) range test and validate AES encryption via transmission log analysis.

Measure device weight and conduct human wear test for 7 days with comfort feedback survey.

Usability test with 10 users setting up device and app; measure completion time and error rate.

Run 72-hour software test cycle to ensure automatic monitoring starts post-setup with no manual actions.

Measure patch thickness and conduct visibility test under clothing on a diverse user group.

Submit adhesive and enclosure for ISO 10993 biocompatibility testing and patch wear irritation test.

Wireless performance test with daily activity simulation and real-time disconnection logging.

Motion artifact evaluation during sitting, walking, and sleeping using annotated ECG signal recordings.

Conduct usability test where users replace battery and adhesive within 1 minute; measure completion time and error rate.

Failure Mode	Effect	Cause
ECG signal noise	Misdiagnosis	Loose electrode connection
BLE transmission fails	Data loss	Interference, low battery
Device adhesive detaches	Interrupted monitoring	Sweat, poor skin contact
Skin irritation	User discomfort	Poor adhesive material
Setup errors	Device not collecting data	Poor user onboarding design
Patch visibility concern	Non-compliance with use	Bulky design
API sync failure	Missed real-time alerts	Backend or app issue
Battery dies early	Device shutdown	High power drain
Lead-off not detected	Inaccurate ECG reading	Software bug

Severity (S)	Occurrence (O)	Detection (D)
9	4	3
8	3	4
7	4	4
5	3	5
8	2	3
4	3	3
7	3	4
9	2	3
10	2	3

RPN	Recommended Action
	Improve electrode placement
108	& shielding
	Add auto-reconnect and alert
96	system
	Redesign adhesive for better
112	grip
	Switch to ISO 10993-certified
75	adhesive
	Simplify app UI, add guided
48	tutorial
	Reduce thickness and use
36	neutral tones
	Add sync verification + alert
84	retry
	Optimize power consumption
54	logic
	Validate detection logic
60	extensively